

6.7 : Continue.

② Integrals involving inverse trig functions.

$$\int \frac{du}{\sqrt{1-u^2}} = \sin^{-1} u + C$$

$$\int \frac{du}{1+u^2} = \tan^{-1} u + C$$

$$\int \frac{du}{u\sqrt{u^2-1}} = \sec^{-1} |u| + C$$

$$\int \frac{du}{a^2+u^2} = \frac{1}{a} \tan^{-1} \frac{u}{a} + C$$

$$\int \frac{du}{\sqrt{a^2-u^2}} = \sin^{-1} \frac{u}{a} + C$$

$$\int \frac{du}{u\sqrt{u^2-a^2}} = \frac{1}{a} \sec^{-1} \left| \frac{u}{a} \right| + C$$

Example : Evaluate

$$A = \int \frac{dx}{1+3x^2}$$

Subs

$$u = \sqrt{3} \cdot x \quad \Rightarrow \quad du = \sqrt{3} \, dx$$

$$\Rightarrow \quad dx = \frac{du}{\sqrt{3}}$$

$$A = \int \frac{1}{1+u^2} \cdot \frac{du}{\sqrt{3}}$$

$$= \frac{1}{\sqrt{3}} \cdot \int \frac{1}{1+u^2} \cdot du$$

$$= \frac{1}{\sqrt{3}} \cdot \tan^{-1} u + C$$

$$= \frac{1}{\sqrt{3}} \cdot \tan^{-1} (\sqrt{3}x) + C$$

